Synopsis

Project Title:Online Purchase Behavior Classification

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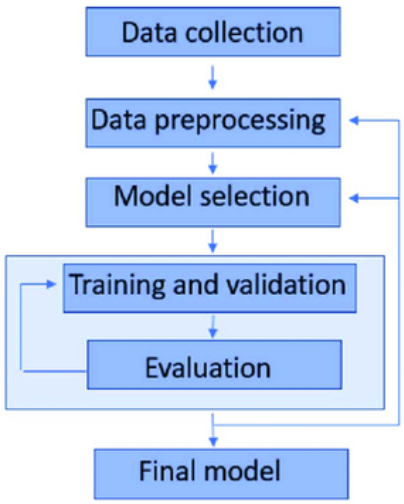
Problem Description:

The objective of this project is to build an AI system that predicts whether a customer will complete a purchase based on their digital footprint. In the modern e-commerce landscape, businesses generate massive amounts of data from user sessions, including **browsing time, clicks, and product views**.

By classifying intent in real-time, companies can personalize marketing, offer timely discounts, and improve user experience. This project uses Logistic Regression and KNN to analyze these behavioral patterns and distinguish between casual browsers and high-intent shoppers.



Steps of Implementation:

The project follows a standard machine learning pipeline to ensure data quality and model reliability:

1. **Data Collection:** Gathering session data such as page views and session duration.
2. **Data Preprocessing:** Cleaning the dataset and **scaling features** to improve model learning.
3. **Feature Selection:** Removing redundant or low-impact behavior data to optimize performance.
4. **Model Selection:** Implementing Logistic Regression and K-Nearest Neighbors **(KNN)**.
5. **Training and Validation:** Using the majority of the data to teach the models and a smaller subset to validate results.
6. **Evaluation:** Measuring success through performance metrics to determine the final model.

Expected Modules / Libraries to be used:

- **Pandas:** Essential for data manipulation and loading the CSV dataset into structured frames.
- **Scikit-Learn:** Provides the core machine learning algorithms for KNN and Logistic Regression, as well as tools for data scaling.
- **Matplotlib / Seaborn:** Used for generating statistical visualizations and plots to understand data distribution.
- **NumPy:** Used for performing high-level mathematical functions on large, multi-dimensional arrays.

Expected Evaluation Metrics and Plots:

To measure the effectiveness of the classification, the following metrics will be used:

- **Accuracy:** To measure the overall percentage of correct predictions.
- **Precision and Recall:** To understand the model's ability to find all shoppers (recall) and how often its "purchase" predictions are correct (precision).
- **F1-Score:** To provide a balance between precision and recall.
- **Confusion Matrix:** A plot used to visualize the true positives vs. false positives for each model.

Required Topics from the Subject:

- **Data Structures (Arrays/Tables):** Necessary for managing the tabular shopping data and feeding it into the ML models.
- **File Handling:** Required for reading the .csv dataset from the local directory into the Python environment.
- **Algorithms:** Understanding the mathematical distance logic behind KNN and the probability functions of Logistic Regression.

Platform Required:

- **Visual Studio Code:** The primary Integrated Development Environment (IDE) for coding and terminal management.
- **UV:** A fast Python package installer and resolver used to manage the project environment.